

CRITICALITY ANALYSIS OF HTR-10 USING THE HIGH-TEMPERATURE GAS-COOLED REACTOR CODE PACKAGE

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ABSTRACT

The 10 MWt High-Temperature Gas-Cooled Reactor-Test module (HTR-10) is a prototype pebble-bed reactor intended to act as a stepping stone for the development of Pebble-Bed Reactor (PBR) technology in China. The International Atomic Energy Agency (IAEA) released a set of benchmark problems that can serve to assess the accuracy of existing codes. These benchmarks support the code-to-code validation of computational models and methods for subsequent safety analysis of future pebble-bed reactors. In this work, initial criticality calculations of the HTR-10 were performed using simplified Serpent Monte Carlo models of the reactor configurations, with consideration of both random, discrete pebble distributions in the pebble bed core. A model was also built in parallel using the HTR Code Package (HCP), a highly-integrated coupling backbone that integrates multiple physics modules, to validate against the Serpent model of the same configuration. The simulation results demonstrated that the HCP code showed good agreement with the Monte Carlo solution in terms of their initial criticality, which indicates HCP's capabilities to accurately model the HTR-10.

KEYWORDS: HTR-10; Pebble-bed reactor; Benchmark analysis; Initial criticality; Serpent; HCP

1. INTRODUCTION

The pebble-bed High Temperature Gas-cooled Reactor (HTGR) is a promising candidate for next-generation nuclear power plants due to its inherent safety features and high temperature capability. The TRISO particles' multi-level barriers, combined with the low power density, result in a significant improvement in safety features, enabling simplified design and reduced costs.

Initiated in 1992, the HTR-10 is a 10 MWt pebble-bed type, graphite-moderated, and helium-cooled modular HTGR operated by the Institute of Nuclear Energy Technology (INET). The reactor first reached criticality in December 2000 and achieved its full power operation in January 2003. During that period, the HTR-10 was used as a tool for the validation of neutronics, and a set of benchmark specifications for its initial critical core experiment by the IAEA [1]. Several studies have been conducted on PBRs, including analyses of its thermal-hydraulic behaviour, fuel behaviour, and safety characteristics. Calculations were performed using a diffusion-based system code known as the Very Superior Old Programs (VSOP), specifically developed for reactor physics simulations with pebble-bed reactors [2]. Jing et al. reported the initial core configuration of the reactor and criticality calculations in 2002 and criticality analyses have been conducted soon after by Colak and Seker [3, 4]. In addition to VSOP, other system codes have been developed for reactor physics simulations with pebble-bed reactors. For instance, the Monte Carlo method has been used for criticality problems in pebble-bed assemblies, including studies dealing with criticality analysis for the HTR-PROTEUS experimental facility using MCNP-4A. [5]. The Multi-Group TINTE (MGT) was later developed for time-dependent reactor dynamics to handle transients and accident simulations for HTGRs while, VSOP is used mainly for core design and reactor operation on a longer time scale [6]. More recently,

the HCP, developed at the Jülich Research Center (FZJ) and RWTH Aachen, facilitates the coupling of these physics modules in a highly integrated manner, allowing for more user-friendly simulation of steady-state and transient operating conditions for pebble-bed reactors [7].

As part one of the coordinated research programmes, IAEA conducts scientific research and analytical evaluations of benchmark problems. In this study, preliminary validation of the HCP is performed by conducting criticality analyses based on transport calculations and comparing the results to the same HTR-10 model built in the Serpent, a Monte Carlo code [8]. These efforts ultimately allow for more accurate safety assessments.

2. DESCRIPTION FOR FIRST CRITICALITY

2.1. Model Description

The HTR-10 reactor has a core vessel that has a cylindrical shape with a cone structure and discharge tube at the bottom. Like any PBR, the core is filled with pebbles, consisting of fuel pebbles and ‘dummy’ pebbles – spheres the same size as fuel pebbles but made only of graphite. The fuel pebbles are billiard-ball-size spheres containing TRISO-coated fuel particles embedded in a matrix. The TRISO particles enclose the UO_2 fuel instead of it being uniformly distributed, creating a double heterogeneity within the pebble-bed reactor. During the operation of the reactor, the pebbles are loaded at the top and gradually move downwards through the core area. The core of the HTR-10 reactor is composed of a cylindrical cavity above a conical zone, known as the “conus”, which directs the pebbles towards a discharge tube. The core, conus, and discharge tube are surrounded by blocks of graphite, known as the reflector, and boronated carbon blocks that have holes for coolant flow, control rods, shutdown absorber balls, and other uses [9].

In the initial criticality configuration, the voids in the core and reflector were filled with ambient air, instead of the helium that is used to cool the reactor during operation. The conus and discharge tube were filled with “dummy” pebbles while the cylindrical core was initially loaded with a mixture of 16,890 fuel pebbles and dummy pebbles in a ratio of 57:43 until the core became critical. The volumetric packing density of the entire core was assumed to be 0.61, which is the widely accepted average value based on empirical data. Based on the pebble diameter, the core diameter, and the assumed packing fraction, it can be confirmed that the number of pebbles used in the initial critical experiment is equivalent to a level core height of 123.06 cm. The reflector region is composed of numerous graphite blocks that are surrounded by boronated carbon bricks at the periphery. Prior to the first criticality experiment, both the reflector and pre-loaded graphite pebbles were dehumidified [9]. For simplicity, the water content is excluded from the reflectors in the calculations.

A fuel pebble is composed of two distinct regions: an outer graphite shell and an inner fuel region. The fuel region contains coated fuel particles that are embedded in a graphite matrix. The fuel particle consists of multiple layers, with the fissile material UO_2 forming the core of the particle. Surrounding the core is a porous carbon buffer layer, followed by a dense inner pyrolytic carbon layer providing a diffusion barrier for fission products, a layer of silicon carbide supplying strength, and finally, an outer pyrolytic carbon layer protecting the SiC layer and impeding diffusion [9]. The nominal data for the fuel and dummy pebble materials, including the coated fuel particles, are listed in Table 1. These parameters and specifications will be used as preliminary benchmarks in Serpent and HCP for the HTR-10 initial criticality analysis.

Table 1: Material properties and dimensions of TRISO particles, fuel and dummy pebbles [9]

Fuel Pebble	Value
Uranium mass per pebble	5.0 g
Density of graphite in matrix and outer shell	1.73 g/cm ³
Uranium enrichment (U^{235} /total U by weight)	17.0 %
Impurities represented by Equivalent Boron Concentration (EBC) in uranium	4.0 ppm
Impurities represented by EBC in graphite	1.3 ppm
Radius of kernel	0.25 cm
Radius of fuel pebble	3.0 cm
Radius of fuel zone in fuel pebble	2.5 cm
Volumetric filling fraction of pebbles in core	0.61
Kernel UO_2 density	10.4 g/cm ³
Coating layers densities, Buffer/PyC/SiC/PyC, respectively	1.1/1.9/3.18/1.9 g/cm ³
Coating thickness, Buffer/PyC/SiC/PyC, respectively	0.09/0.04/0.035/0.04 cm
Average number of particles per pebble	8335
Dummy Pebble (Original Benchmark)	Value
Density of graphite	1.73 g/cm ³
Impurities represented by EBC in graphite	1.3 ppm
Radius of dummy pebble	3.0 cm
Dummy Pebble (Deviated Benchmark)	Value
Density of graphite	1.84 g/cm ³
Impurities represented by EBC in graphite	0.125 ppm
Radius of dummy pebble	3.0 cm

2.2. Benchmark Problems Specifications

The benchmark model used for the critical configuration of HTR-10 is the simplified model, where the upper layer of the core is approximated as a flat plane and the reflectors are made azimuthally symmetric. Furthermore, the coolant flow, control rods and small absorber balls (KLAK) channels that are present in the high-fidelity model are azimuthally homogenised and replaced with structural materials. The total number of fuel and dummy pebbles remains unchanged at 9627 and 7263, respectively, with each pebble containing 8335 coated fuel particles. The simplified model makes certain adjustments to accommodate the limitations of codes that have less sophisticated geometric modelling capabilities. In the benchmark configurations used in this study, the region between the top reflector and top of the pebble bed core was filled with a void space, in contrast to the IAEA document's benchmark configuration, where this region is filled with either helium or air [1].

The original B1 benchmark problem involves determining the amount of loading for the first criticality with the core filled with helium and core temperature of 27 °C, without the insertion of control rods or shutdown absorbers [1]. The initial criticality experiment reported a loading height of 123.06 cm from the upper surface of the conus region and thus, this height is used for both the Serpent and HCP model. The core temperature refers to the temperature of the fuel pebbles and the

surrounding reflectors encompassed within the core physics model, as described in the HTR-10 core configuration.

The original B2 benchmark problem involves calculating the effective multiplication factor (k_{eff}) of the full core at three different core temperatures: 27 °C (known as B21), 120 °C (known as B22) and 250 °C (known as B23), without the insertion of control rods or shutdown absorbers [1]. This problem is designed to investigate the temperature dependence of k_{eff} . This benchmark employs the same model as the B1 benchmark, but with a taller pebble height of 180.12 cm for the full-core analysis, in contrast to the 123.06 cm height used for the initial core criticality analysis. Due to the taller pebble bed height of 180.12 cm used for full-core criticality analysis in the B2 benchmark, the total number of pebbles increases to 24721, achieving a packing fraction of 0.61. This total consists of 14091 fuel pebbles and 10630 dummy pebbles, where each fuel pebble still contains 8,335 coated fuel particles.

The original benchmark was established prior to achieving the actual first core criticality, and it has been modified to reflect the actual experimental configuration, thus creating a revised version also known as the deviated benchmark [1]. The core temperature reported was 15 °C and the critical core height remained at 123.06 cm. Compared to the original benchmark specifications, the changes made in the deviated benchmark is as follows:

- (a) The density of the dummy pebbles increased from 1.73 g/cm³ to 1.86 g/cm³.
- (b) The impurities represented by EBC in graphite of the dummy pebbles decreased from 1.3 ppm to 0.125 ppm.
- (c) The core is filled with ambient air.

3. MODELLING METHODOLOGY

The complex geometry in pebble bed reactors due to the randomly distributed spherical fuel and dummy elements, makes neutron transport in the system difficult to accurately model. Discrete Element Method (DEM) is a computational technique that allows the modelling of the packing structure and interactions between individual pebbles, which explicitly captures the double heterogeneity of the system. The positions of the pebbles are first determined using a DEM code called LAMMPS and these discrete coordinates, together with the radius and pebble type, are then used to construct the MC model with Serpent [10]. The model boundaries are modelled in LAMMPS according to the cylindrical core specifications and the pebble bed is generated by randomly pouring the pebbles into the model under the influence of gravitational force and contact forces. The pebble distribution evolves according to the Newton's second law which is defined through a series of dimensionless parameters such as friction coefficient and normal elastic constant. After the pebbles have settled, the critical height and number ratio of fuel and dummy pebbles were achieved using a simple post-processing Python script, while ensuring a packing fraction of 0.61 for the critical core configuration. On the other hand, the random positions of TRISO particles in the graphite matrix of a fuel pebble are generated with Serpent's particle automated disperser routine, which produces a distribution file that can be used as an input card. Figure 1 shows the packed pebble bed generated with LAMMPS, together with its radial distribution function. In this case, the radial distribution function, $g(r)$ shows how the density varies as a function of the pair separation distance to a particular pebble. The graph in Figure 1 (b) recorded a peak at a distance 6 cm, which is due the surrounding pebbles in direct contact.

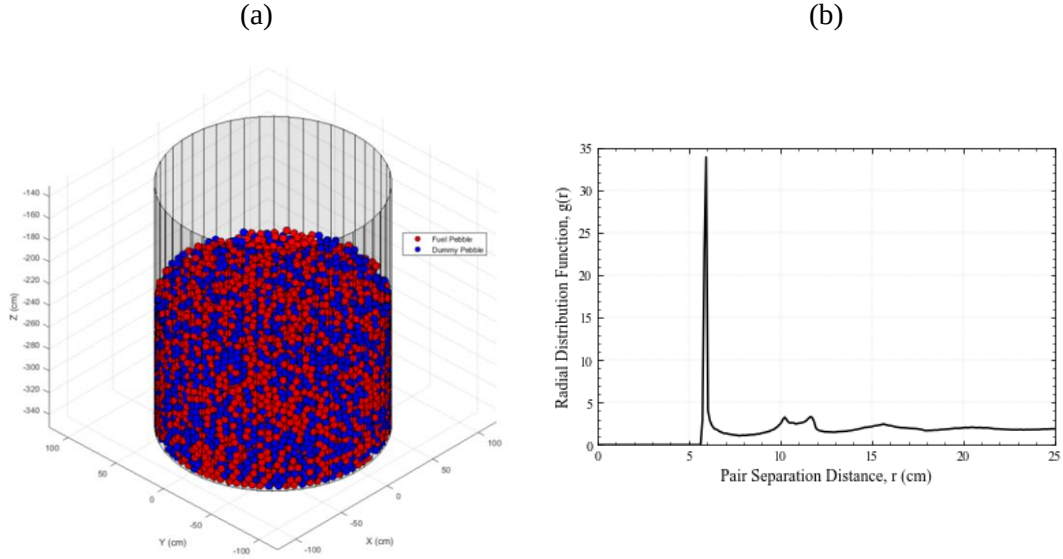


Figure 1: (a) The distribution of the mixed pebbles for the critical core configuration generated in LAMMPS. The fuel and dummy pebbles are coloured in red and blue, respectively. (b) The radial distribution function (RDF) graph of the packed pebble bed.

The present study involved performing neutronics calculations using the 3D continuous-energy Monte Carlo reactor physics code Serpent 2, developed at VTT Technical Research Centre [8]. Serpent was selected for this study because it provides the functionality to create an explicit stochastic geometry model that enables the TRISO particles and fuel pebbles to be defined explicitly in stochastic configurations. The explicit geometry model works on multiple levels, including the particles within a pebble and the pebbles within the core, as depicted in Figure 2. This model has been verified in realistic double heterogeneous reactor setups. Moreover, it uses delta-tracking methods to simulate the transport of neutrons which significantly improve the computational performance in geometries. These reasons make Serpent an excellent choice to model double heterogeneous geometries, such as the HTR-10 reactor, with increased accuracy. All the simulations in Serpent were performed using the ENDF/B-VII.0 cross section data library, which is consistent with the library used in HCP [11].

The geometry and material compositions of the simplified HTR-10 benchmark model were based on a technical document provided by the IAEA [9]. The geometric cells provided in the document are given in Figure 3 (a). In the initial criticality model, the core was filled with a pebble mixture generated with LAMMPS while the lower conus and discharge tube region were filled with dummy pebbles arranged in a Body-Centred Cubic lattice. The lattice in the conus region was adjusted to achieved a packing fraction of approximately 0.61. The region between the top reflector and top of the pebble bed core was filled with a void space, which is consistent with the model built in HCP, while the spaces between the pebbles were filled with either air or helium. The reflector region of the model consists of multiple graphite blocks, which are enclosed by boronated carbon bricks around the edges. The axial cross section of the model and the radial cross section at the height of 300 cm from the top is shown in Figure 3 (b).

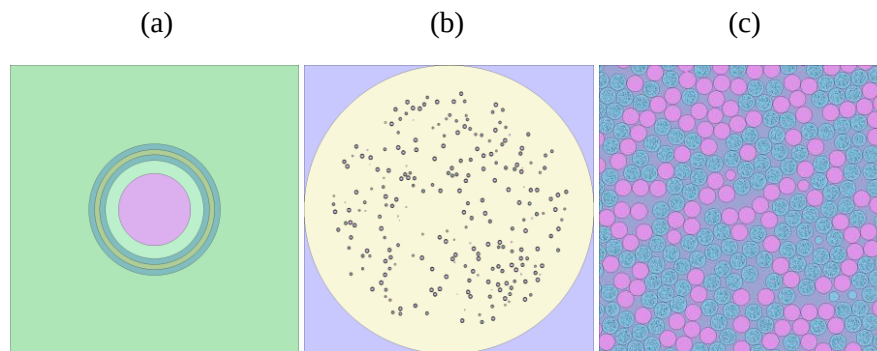


Figure 2: (a) Infinite grain model (TRISO particle), (b) infinite fuel pebble model with random distribution of TRISO particles, and (c) mixture of fuel and dummy pebbles (represented in blue and pink, respectively) in a packed pebble bed

(a)

(b)

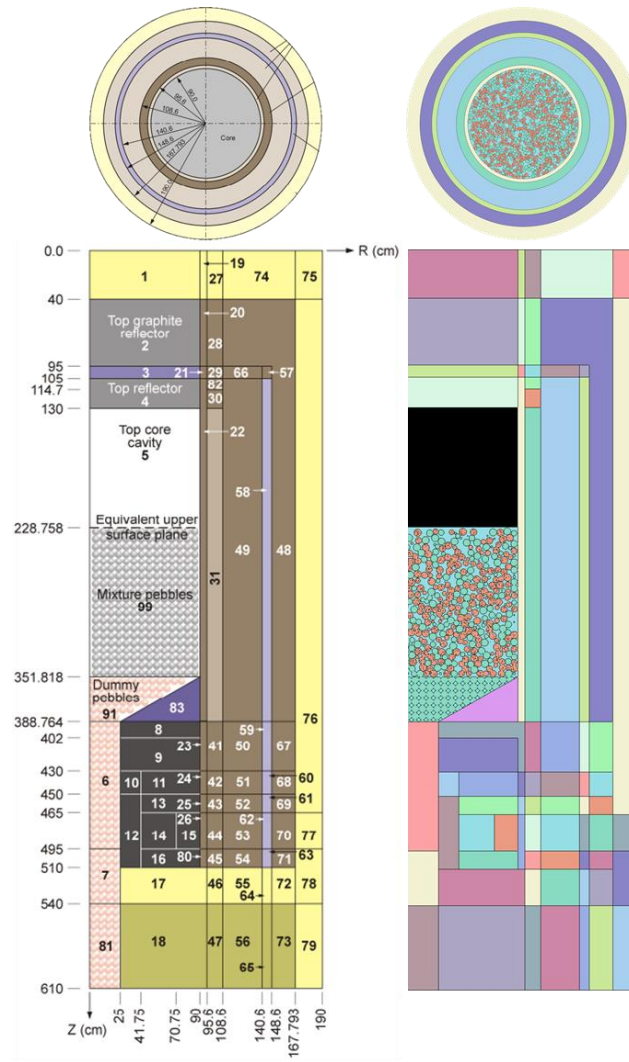


Figure 3: Simplified HTR-10 (a) benchmark model and (b) Serpent model

An exact replica of the model built in Serpent was created in HCP, utilising the TRISHA spectrum code solver and the Bende's Dancoff factor method [7]. TRISHA utilises a new data library generator, which helps the user generate nuclear cross section data with full control. Unit cells used for the TRISO particles and fuel pebbles are automatically updated to account for the spatial and time dependent nuclide concentrations, which eliminates the need to manually define different unit cells as input [12]. It combines a 0D diffusion solver and a 0D transport code based on the B_1 method [13]. An analytical method to calculate the Dancoff factor that is incorporated in the code uses first-flight escape and transmission probabilities while the other method is based on the chord method [14, 15]. Both methods have been validated with Monte Carlo calculations and are effective in handling the double heterogeneity associated with pebble bed reactors [7].

The MGT-3D code was responsible in handling HTGR transient and accident simulations, and integrating it into HCP involves decoupling the MGT Fluid Dynamics (MGT-FD) and MGT Neutronics (MGT-N) modules [6]. With the MGT-N module re-implemented into the HCP framework, calculations of the spatial neutron flux and fission power distribution becomes possible. The MGT-N code employs a finite-difference and leakage iteration method, and is capable of modelling steady-state and transient behaviour of a 3D reactor system represented in cylindrical

coordinates [16]. All the modules in HCP are linked by a newly developed backbone that enables integrated and optimised management of the code system's inputs and outputs [7]. In this work, three input files that deals with different aspects of the simulation in HCP are created to model the HTR-10 based on the benchmark problems. The node grids within a shared volume used for fuel management in the core are shown in Figure 4.

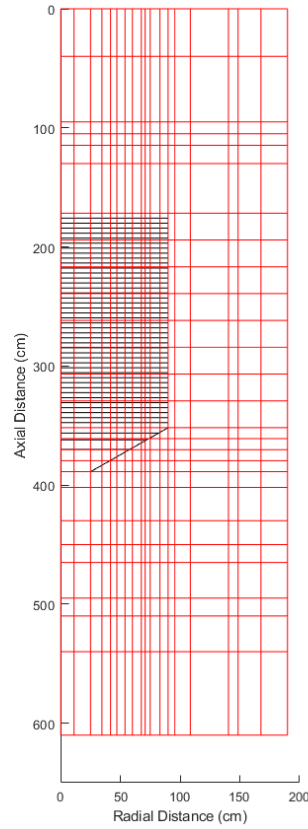


Figure 4: Schematic of the volume shared between the mesh grid (red) and node grid (black)

4. RESULTS

A comparison of the calculated k_{eff} using Serpent and HCP for both original and deviated benchmark of the HTR-10 simplified model is presented in Table 2; Serpent was used as the reference solution. The neutron population in Serpent was defined to be 5000 neutrons per cycle and the statistical uncertainty gave a range of ± 0.00080 to ± 0.00089 .

Table 2: Comparison of the calculated k_{eff} using Serpent and HCP

Benchmark Problem		Original Benchmark			Deviated Benchmark		
		Serpent	HCP	Relative Error (pcm)	Serpent	HCP	Relative Error (pcm)
B1		1.01474	1.01536	-61.1	1.02415	1.02446	-30.3
B2	B21	1.14401	1.14822	-368.0	1.15246	1.15670	-367.9
	B22	1.13262	1.13363	-89.2	1.14146	1.14171	-21.9
	B23	1.11882	1.11686	175.2	1.12731	1.12468	233.3

The relative differences between the calculated k_{eff} values obtained from Serpent and HCP falls below 400 pcm for both the B1 and B2 benchmark problem. Therefore, this indicates a good agreement between the two codes and more importantly, the results demonstrate the accuracy of the HCP code in modelling the HTR-10 configuration.

Although both the Monte Carlo and deterministic codes showed good consistency, a possible reason for the difference between the calculated k_{eff} values could be the hexagonal regular lattice pebble distribution in the conus region of the Serpent model, which does not result in an exact packing fraction of 0.61 defined in HCP. In addition, the discrepancies in the initial criticality observed between the experimental and computational models could be attributed to uncertainties in the water content of the graphite core.

HCP has been implemented with the aim of achieving a well-balanced trade-off between accuracy and computational burden while ensuring user-friendliness by making use of XML files for the user input. HCP demonstrates significant advantages over Serpent when considering the same benchmark problem. On the same machine, HCP requires 3 times less memory and runs 80 times faster than Serpent. It is important to acknowledge that a direct comparison between a Monte Carlo code and a deterministic one may not be entirely fair, particularly in the absence of uncertainty analyses for the deterministic code. Nonetheless, HCP offers superior computational efficiency and faster simulations. However, HCP does have certain limitations when it comes to modelling high fidelity systems, such as accurately representing the azimuthal configuration of control rod borings and coolant flow regions.

5. CONCLUSION

A simplified HTR-10 model was built using the same input configuration as for the first criticality calculation using Serpent and HCP code. The Serpent model incorporated a stochastic discrete arrangement of TRISO particles within the fuel pebbles and a random distribution of pebbles within the core of the pebble bed using DEM code LAMMPS.

The comparison of k_{eff} values between the simplified models of Serpent and HCP for the initial critical core configuration shows an outstanding agreement, indicating the accuracy of the HCP code. This study suggests that HCP is a reliable tool for modelling high-temperature gas-cooled reactors, such as the HTR-10. It could be used in future studies to investigate other reactor designs, such as HTR-PM or operating conditions. Lastly, the study of neutron spectra and spatial power distributions in future analysis will provide a more comprehensive evaluation of the performance of both Serpent 2 and HCP.

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